Transaction Failure Analysis & Expected Discoveries

- **Scenario:** Customer tries to transfer ₹50,000. Transaction failed. Error shown: *"Transaction could not be completed."*
- **Objective:** Perform a root-cause analysis (RCA) and identify potential validation or system failures behind the generic error message.

Breakdown of the 9 Expected Discoveries (Troubleshooting & Test Scenarios)

1. **Daily Limit Exceeded**
    - *Investigation / Test Case:* Check if the user's account has a pre-configured daily transaction limit (e.g., ₹25,000 or ₹50,000) that has already been reached or crossed with this single transaction.
    - *Expected System Behavior:* The app should ideally display a specific error like *"Daily transfer limit exceeded"* instead of a generic failure message.
2. **Insufficient Balance**
    - *Investigation / Test Case:* Verify the account's ledger and available balance. Ensure that the total amount debited (Transfer amount ₹50,000 + applicable transfer fees/charges) does not exceed the available balance.
    - *Expected System Behavior:* Prompt the user with *"Insufficient funds in your account."*
3. **Bank Server Unavailable**
    - *Investigation / Test Case:* Check the availability and health status of the destination bank's server or the originating bank's core processing gateway.
    - *Expected System Behavior:* Handle HTTP 503 errors gracefully with a message like *"Bank server is currently down. Please try again later."*
4. **OTP Expired**
    - *Investigation / Test Case:* Verify if the One-Time Password (OTP) entered by the user took longer than the validity window (e.g., 2 minutes) or if the user entered an old/stale OTP.
    - *Expected System Behavior:* Display *"OTP expired. Please request a new OTP."*
5. **Network Timeout**
    - *Investigation / Test Case:* Simulate packet loss, slow internet connection, or an interrupted client-server connection during the API request payload transmission.
    - *Expected System Behavior:* Trigger a network timeout exception handler and prompt *"Connection timed out. Please check your network."*

6. **Beneficiary Inactive**
    - *Investigation / Test Case:* Check the status of the payee/beneficiary account. The beneficiary might have been frozen, closed, or deactivated by the bank.
    - *Expected System Behavior:* Flag the account status and show *"The recipient account is inactive or blocked."*
7. **Fraud Rule Triggered**
    - *Investigation / Test Case:* Evaluate risk management systems. A sudden high-value transfer of ₹50,000 to a newly added beneficiary might trigger an automated anti-fraud security block.
    - *Expected System Behavior:* Temporarily hold the transaction for security verification or display *"Transaction flagged for security review."*
8. **Core Banking Unavailable (CBS Down)**
    - *Investigation / Test Case:* Check if the backend Core Banking System (CBS) database is undergoing maintenance or experiencing an outage, preventing account updates.
    - *Expected System Behavior:* Return a system-level failure notice or fallback maintenance page.
9. **UPI Switch Unavailable**
    - *Investigation / Test Case:* If the transfer is routed through UPI, check the national UPI switch status (NPCI gateway down/latency issues).
    - *Expected System Behavior:* Display *"UPI service is temporarily unavailable. Try using IMPS or NEFT."*